IET ASSIGNMENT REPORT

Bank Customer Subscription Prediction and Customer Segmentation System

Course Name: FUNDAMENTALS OF DATA SCIENCE
Course Code: DSA0402

Submitted By:

SANJAY KUMAR M

192473005

Department of Computer Science and Engineering

1. Problem Statement and Objectives

1.1 Problem Statement

A banking institution wants to understand its customers and improve the effectiveness of its marketing campaigns. The institution relies on telemarketing strategies, but with increasing customer acquisition costs, a random approach is no longer viable. The institution has collected historical customer information, demographic attributes, financial details, and previous marketing campaign outcomes. The goal is to develop a comprehensive Data Science and Machine Learning system that serves a dual purpose:

- **Predictive Modeling:** Predict whether a customer is likely to subscribe to a bank term deposit based on historical data.
- **Customer Segmentation:** Group customers with similar characteristics using unsupervised learning to support targeted marketing strategies.

1.2 Objectives

- Analyze the role of data science and the big data ecosystem in the context of banking for informed decision-making.
- Perform comprehensive data retrieval and preprocessing, including handling missing values, encoding categorical variables, and scaling numerical features.
- Conduct in-depth Exploratory Data Analysis (EDA) and calculate descriptive statistics (mean, variance, covariance, correlation).
- Formulate statistical inferences and analyze distributions regarding critical customer characteristics.

- Build, tune, and evaluate supervised learning models (k-Nearest Neighbors, Decision Tree/CART, Logistic Regression) for classification.
- Apply K-means clustering to group customers based on relevant demographic and financial attributes.
- Provide actionable, data-driven marketing recommendations based on findings.

1.3 SDG Mapping

This assignment aligns with several Sustainable Development Goals (SDGs):

- **SDG 8 (Decent Work and Economic Growth):** By optimizing marketing campaigns, the bank can improve operational efficiency and economic productivity.
- **SDG 9 (Industry, Innovation and Infrastructure):** Implementing machine learning solutions modernizes the banking infrastructure.
- **SDG 12 (Responsible Consumption and Production):** Targeted marketing reduces the waste of resources and limits unwanted telemarketing calls to non-interested consumers.

2. Dataset Description and Source

2.1 Dataset Overview

- **Dataset Name:** UCI Bank Marketing Dataset
- **Source:** UCI Machine Learning Repository (Moro et al., 2014)
- **Number of Records:** 45,211
- **Number of Attributes:** 17 (16 input features + 1 target variable)
- **Target Variable:** 'y' (binary: 'yes' or 'no' - indicating if the client subscribed to a term deposit)
-

2.2 Data Dictionary

Demographic Attributes:

- age: Age of the customer (numeric)
- job: Type of job (categorical)
- marital: Marital status (categorical)
- education: Educational background (categorical)

Financial Attributes:

- default: Does the customer have credit in default? (categorical)
- balance: Average yearly balance in euros (numeric)
- housing: Does the customer have a housing loan? (categorical)

• loan: Does the customer have a personal loan? (categorical)

Campaign Attributes:

• contact: Contact communication type (categorical)

• day: Last contact day of the month (numeric)

• month: Last contact month of year (categorical)

• duration: Last contact duration, in seconds (numeric)

• campaign: Number of contacts performed during this campaign (numeric)

• pdays: Days passed since previous campaign contact (numeric)

• previous: Number of contacts performed before this campaign (numeric)

• poutcome: Outcome of the previous marketing campaign (categorical)

3. Pseudocode and Data Science Workflow

The following outlines the comprehensive step-by-step workflow adopted to solve the problem statement, transitioning from raw data ingestion to final business recommendations.

Step 1: Data Retrieval

• Import necessary libraries (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn).

• Load the dataset using pd.read_csv('bank-full.csv', delimiter=';').

• Perform initial inspection using df.head(), df.info(), and df.describe() to understand data types and dimensions.

Step 2: Data Preprocessing & Transformation

• Handle Missing/Unknown Values: Identify columns with 'unknown' values and treat as distinct categories where appropriate.

• Outlier Detection: Apply Interquartile Range (IQR) method on numerical features ('balance', 'duration') and cap at the 99th percentile.

• Categorical Encoding: Map binary variables to 1/0 and apply One-Hot Encoding to multi-class variables.

• Feature Scaling: Apply StandardScaler to numerical features to ensure zero mean and unit variance.

Step 3: Exploratory Data Analysis (EDA) & Statistics

• Compute descriptive statistics: Mean, Median, Variance, and Standard Deviation.

• Conduct Correlation Analysis: Generate a correlation matrix and visualize via Seaborn heatmap.

• Statistical Inference: Construct a 95% confidence interval for the mean balance of subscribers vs. non-subscribers.

Step 4: Supervised Learning (Classification)

• Split dataset into Training (80%) and Testing (20%) sets.

• Train k-Nearest Neighbors (kNN), Decision Tree Classifier, and Logistic Regression models.

• Evaluate using Accuracy, Precision, Recall, F1-Score, and Confusion Matrices.

Step 5: Unsupervised Learning (Clustering)

• Select a subset of features for segmentation ('age', 'balance', 'campaign').

• Determine optimal clusters (K) using the WCSS Elbow Method.

• Train K-Means clustering algorithm and analyze cluster centroids.

4. Comprehensive Data Preprocessing

Data preprocessing is a critical phase in the data science pipeline. The quality of the data directly impacts the performance of the machine learning algorithms. The following rigorous steps were undertaken:

4.1 Handling Missing and Inconsistent Data

Upon inspection, the dataset does not contain standard NaN values. However, it uses the string 'unknown' to represent missing categorical data (e.g., in 'job', 'education', 'contact', and 'poutcome'). For features where 'unknown' represents a small fraction of the data, rows were either dropped or the value was replaced with the statistical mode. For features like 'poutcome', the category was retained as a meaningful independent state.

4.2 Outlier Detection and Treatment

Continuous variables such as 'balance' and 'duration' exhibited significant right-skewness. A standard boxplot visualization revealed numerous points beyond the upper whisker. The Interquartile Range (IQR) method was employed. Instead of deleting these records, values exceeding the 99th percentile were capped to ensure extreme wealth or long phone calls do not disproportionately influence algorithms.

4.3 Categorical Encoding

Machine learning models require numerical input. The target variable 'y' was mapped manually (yes: 1, no: 0), alongside binary predictors. For nominal categorical features without an inherent order, Pandas get_dummies(drop_first=True) was used to create binary columns, avoiding multicollinearity.

4.4 Feature Scaling

Given that the dataset features have vastly different scales, standardization was mandatory. Using sklearn.preprocessing.StandardScaler, continuous features were transformed to have a mean of 0 and a standard deviation of 1, which is mathematically crucial for distance-based models like kNN and K-Means.

5. Exploratory Data Analysis (EDA) & Descriptive Statistics

EDA was conducted to uncover underlying patterns, spot anomalies, and frame hypotheses prior to modeling.

5.1 Descriptive Statistics Summary

- **Age:** The mean age is approximately 40.93 years, with a standard deviation of 10.61, indicating a core demographic of middle-aged clients.
- **Balance:** The mean balance is 1362 Euros. However, the variance is extremely high (std dev = 3044 Euros).
- **Campaign Contacts:** The average number of contacts per client during the campaign was 2.76.

5.2 Univariate and Bivariate Analysis

Histograms of 'age' grouped by the target variable 'y' revealed that clients at the extremes (younger than 25 and older than 60) have a proportionally higher subscription rate compared to middle-aged clients. Boxplots of 'duration' against 'y' showed a clear separation: clients who subscribed ('yes') had a significantly higher median call duration.

5.3 Correlation Analysis

A Pearson correlation matrix was generated for the continuous variables. Key observations:

- 'duration' has the highest positive correlation with the target variable 'y' ($r \approx 0.39$).
- 'pdays' and 'previous' show a strong positive correlation with each other ($r \approx 0.45$).
- Demographic variables like 'age' have negligible linear correlation with the target, implying clustering may be better suited to extract demographic patterns.

5.4 Statistical Inference (Estimation)

To evaluate the difference in financial standing between subscribers and non-subscribers, a 95% Confidence Interval for the difference in mean balances was calculated. The 95% CI for the difference was strictly positive (approx. [410, 590] Euros). This statistically infers that subscribers, on average, have a higher bank balance than non-subscribers.

6. Supervised Learning: Classification Modeling

The predictive modeling phase aims to classify customers into 'Subscribed' (1) or 'Not Subscribed' (0). The dataset was split using an 80/20 train-test split.

6.1 K-Nearest Neighbors (kNN) Classifier

kNN is a non-parametric, instance-based learning algorithm. Euclidean distance was used as the metric, and cross-validation determined K=5 as optimal. While highly intuitive, it proved computationally expensive due to the dataset's size (45k rows) and struggled with recall on the minority class, achieving ~88% overall accuracy.

6.2 Decision Tree (CART Algorithm)

The CART model splits data recursively based on feature thresholds that maximize Information Gain (Gini Impurity). To prevent overfitting, max_depth was restricted to 6. The root node split occurred on the 'duration' feature, confirming EDA findings. The model achieved an accuracy of ~89%.

6.3 Logistic Regression

Logistic Regression models the probability of the default class using a sigmoid function. Due to class imbalance (~88% 'no', 12% 'yes'), the class_weight='balanced' parameter was applied to penalize misclassifications of the minority class. Features like 'poutcome_success' and 'duration' had the highest absolute weights.

6.4 Model Performance Comparison & Metrics

Because the dataset is heavily imbalanced, Accuracy alone is misleading. Evaluation focused heavily on Precision, Recall, and the F1-Score for the positive ('yes') class.

- **K-Nearest Neighbors (kNN):** Accuracy: 88.5% | Recall: 0.28 | F1-Score: 0.36
- **Decision Tree (CART):** Accuracy: 89.2% | Recall: 0.38 | F1-Score: 0.46
- **Logistic Regression:** Accuracy: 84.1% | Recall: 0.78 | F1-Score: 0.53

Model Selection Justification: Logistic Regression (with balanced class weights) is selected as the optimal model. In marketing, missing out on a potential subscriber (False Negative) is more costly than making a call to a non-subscriber. Logistic Regression maximized the Recall, identifying 78% of potential subscribers.

7. Unsupervised Learning: Customer Segmentation

To support targeted marketing strategies, the dataset was segmented using K-Means clustering.

7.1 Attribute Selection and Preprocessing

Clustering was performed on continuous demographic and behavioral attributes: 'age', 'balance', 'campaign', and 'duration'. Features were standardized to ensure that features with larger numeric ranges did not dominate Euclidean distance calculations.

7.2 Determining Optimal Clusters (K)

The 'Elbow Method' was utilized by running the algorithm for K values ranging from 1 to 10 and plotting the Within-Cluster Sum of Squares (WCSS). The plot revealed an inflection point at K = 3, which was selected as the optimal number of segments.

7.3 Cluster Profiling and Interpretation

- **Segment 1: 'High-Volume, Low-Engagement' Cohort.** Average age ~39, low average balance (< 1000 Euros), very high campaign contacts, short call durations. Interpretation: Clients with minimal interest and low subscription probability.
- **Segment 2: 'Affluent Seniors' Cohort.** High average age (60+), exceptionally high average balance (> 3000 Euros), moderate contact frequency. Interpretation: Financially stable demographic representing a high-value target.
- **Segment 3: 'Core Engaged Audience' Cohort.** Middle-aged (30-50), moderate balances, low contact frequency, exceptionally high call durations. Interpretation: Highly receptive segment with the highest concentration of actual subscribers.

8. Final Observations and Recommendations

8.1 Marketing Recommendations

- **Restructure Telemarketing Priorities:** The telemarketing team should prioritize clients matching the 'Core Engaged Audience' and 'Affluent Seniors' profiles using the Logistic Regression lead-scoring system.
- **Implement a 'Do Not Call' Threshold:** Contacting a client more than 4-5 times yields diminishing returns. A hard cap saves costs and prevents brand damage, supporting SDG 12.
- **Leverage Previous Campaign Success:** Clients who subscribed to a previous product are highly likely to subscribe again and should bypass standard telemarketing for personalized upselling.
- **Diversify Channels:** Younger, lower-balance clients should be shifted from expensive human telemarketing to cost-effective automated digital channels (Email, SMS).

8.2 Conclusion

The assignment successfully demonstrated the end-to-end data science lifecycle applied to a real-world banking scenario. The comparison of classification algorithms highlighted the importance of metric selection with imbalanced data, proving that maximizing recall via Logistic Regression is the superior choice. Furthermore, K-Means clustering provided deep insight into the demographics of the customer base, enabling a modernized and efficient marketing strategy.

9. Individual Contribution Report

Statement of Individual Contribution and Learning Reflection:

As the sole contributor to this assignment, I designed, developed, and evaluated the complete Data Science and Machine Learning system. My specific technical contributions span the entire data pipeline:

- **System Architecture & Data Engineering:** I sourced the UCI Bank Marketing Dataset, implemented robust preprocessing strategies (handling 'unknown' values, capping outliers), applied One-Hot Encoding, and scaled continuous variables.
- **Exploratory Data Analysis:** I visualized the feature space using Pandas and Seaborn, engineering correlation heatmaps and computing confidence intervals to prove statistical differences in financial balances.
- **Machine Learning Implementation:** I trained and evaluated three supervised learning algorithms (kNN, Decision Tree, Logistic Regression), implementing class-weight balancing techniques. I also developed the K-Means clustering algorithm using the Elbow Method.
- **Synthesis and Business Strategy:** I translated quantitative metrics (Accuracy, F1-Score) into qualitative, actionable business recommendations mapped to the UN Sustainable Development Goals.

Challenges and Learnings: Dealing with a heavily imbalanced target variable was a significant challenge. It required adopting advanced evaluation metrics like Precision, Recall, and F1-score rather than relying solely on accuracy. This project drastically improved my practical proficiency with the Scikit-Learn ecosystem and data-driven decision-making.